

Week 1 Interim Submission – Ansarah Mohammed

AI-Assisted Emergency Department Triage Research

Literature Review Notes

Paper 1: Tyler et al. (2024) – Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

This scoping review looked at how artificial intelligence and machine learning are being used in emergency department triage. The authors reviewed peer-reviewed studies where AI was used in ED triage and where patient outcomes were reported. They found that machine learning models often performed better than conventional triage systems for predicting risk, hospital admission, disease identification, and resource needs. However, the review also showed that many studies still have issues with retrospective data, single-site datasets, ethics, provider acceptance, and long-term patient outcomes.

Paper 2: Da'Costa et al. (2025) – AI-Driven Triage in Emergency Departments: A Review of Benefits, Challenges, and Future Directions

This review focused on overcrowding, limited resources, and inconsistent patient prioritisation in emergency departments. The authors reviewed literature on AI triage systems, machine learning, natural language processing, real-time analytics, and ethical concerns. They found that AI-driven triage may improve patient prioritisation, reduce waiting times, support staff, and improve resource allocation. However, they also identified challenges such as poor data quality, algorithmic bias, lack of clinician trust, privacy concerns, accountability, and the need for more diverse datasets.

Paper 3: Araouchi & Adda (2024) – TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

This paper focused on emergency department overcrowding and the pressure placed on triage staff when patient numbers and case complexity increase. The authors tested several machine learning models, including SVM, Random Forest, ANN, GBM, Logistic Regression, XGBoost, and a stacking model, to predict triage levels using the Korean Triage and Acuity Scale. The stacking model performed best, with 80.05% accuracy, 80.27% precision, 73.26% recall, and a 74.41% F1-

score. However, the authors stated that the model relied on a single dataset, which may limit its generalisability to hospitals with different patient groups, seasonal patterns, and clinical settings.

Paper 4: Kang et al. (2025) – Development of Predictive Model for Patient Triage at Emergency Department

This paper focused on assigning emergency department patients to the correct care zone quickly and safely. The authors used data from Sengkang General Hospital in Singapore and trained machine learning models using triage information such as vital signs, demographics, NEWS scores, and nurses' triage notes. The Light Gradient-Boosting Machine model performed best, with an AUC of 0.95 on the holdout test set and 0.94 on a later clinical validation set. However, the model depended on local data and nurses' notes, had some difficulty classifying younger patients, and would need live testing, more diverse data, and regular updating before wider use.

Paper 5: Alomari et al. (2025) – Safety and Accuracy of AI in Triage Patients in the Emergency Department

This paper examined whether ChatGPT could safely support emergency department triage using the Canadian Triage and Acuity Scale. The authors conducted a prospective observational study with 138 patients, comparing triage scores from ED physicians, ChatGPT, and senior ED consultants. ChatGPT had high agreement with ED physicians at 85.61%, but its accuracy compared with the senior ED consultant was only 42.86%. The authors stated that the study was limited by its small sample size, the use of a general AI model not trained specifically for medical triage, and possible effects from case complexity, missing data, and physician experience.

Paper 6: Wang et al. (2025) – Patient Triage and Guidance in Emergency Departments Using Large Language Models

This paper studied whether large language models such as ChatGPT could support ED triage and guide patients to the correct department. The authors tested GPT-4o and GPT-4-Turbo using MEWS-based triage scenarios, then tested GPT-4o for outpatient department guidance. After prompt engineering, GPT-4-Turbo achieved 100% triage accuracy, while GPT-4o achieved 96.20%; GPT-4o also achieved 92.63% accuracy for department guidance. However, the study used Chinese patient scenarios, Chinese department structures, and structured descriptions, so the results may not fully reflect real-world emergency departments in other healthcare systems.

Background Source: Yancey & O'Rourke (2023) – Emergency Department Triage

This source provided background information on modern triage methods used in emergency care. It explained that triage sorts and prioritises patients based on severity, urgency, and available resources. It also described systems such as START, Jump-START, the Emergency Severity Index, the Australasian Triage Scale, the Canadian Triage and Acuity Scale, and the Manchester Triage System. This was useful background because it showed that current triage is already structured, but still depends on clinical judgement, vital signs, symptoms, local protocols, and available resources.

Two Feasible Literature Gaps

Gap 1: Limited validation in Caribbean emergency department settings

Several papers showed promising AI triage performance, but many models were tested using data from one country, one hospital, or one specific triage system. For example, Araouchi & Adda used KTAS data, Kang et al. used Singapore hospital data, and Wang et al. used Chinese patient scenarios. This matters for Mercer because Caribbean patients, hospital resources, triage workflows, and seasonal patient patterns may differ from those settings. A 12-week pilot could address this by testing an AI triage support model on de-identified Mercer ED data and comparing its performance with existing triage outcomes.

Gap 2: Limited evidence on how AI can fit into existing triage workflow

The papers show that AI may support triage, but they do not fully show how doctors and nurses would use it in daily ED work without adding confusion or extra workload. This is important because AI should not replace clinical judgement or create a completely new process that slows down staff. A 12-week pilot could test AI as a support layer beside existing triage methods, focusing on whether it helps identify high-risk patients while keeping the workflow practical for doctors and nurses.

Draft 90-Word Problem Statement

At Mercer General Hospital, nurses must make quick triage decisions during high traffic in the emergency department, with limited resources available. Current AI triage studies show useful results, but most models were tested outside Caribbean patients, hospitals, and workflows, and few show how AI can fit smoothly beside existing triage methods. This 12-week pilot will test whether a machine-learning support tool using de-identified Mercer ED data can improve high-risk patient identification without disrupting doctors' and nurses' usual workflow.

Word count: 82 words

AI Tool Use

I used ChatGPT to help organise and shorten my paper summaries after reading the papers myself. I used it to check whether each summary included the problem, method, outcome, and stated limitation. I also used it to help revise my problem statement so that it was under 90 words and written in clearer British English. I did not use AI to invent sources, and I checked the paper titles, DOIs, and main findings against the papers provided.

References

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2. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*. 2025;197:105838. doi:10.1016/j.ijmedinf.2025.105838.
3. Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science*. 2024;251:430–437. doi:10.1016/j.procs.2024.11.130.
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